Mohamed Helmy

Android & Kotlin Multiplatform Engineer

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SUMMARY

Android and Kotlin Multiplatform engineer with hands-on experience shipping cross-platform apps and libraries. Specialized in Jetpack Compose, modular architecture, offline-first sync, and secure mobile systems. Proven ability to modernize legacy codebases, lead technical initiatives, and deliver production-ready solutions at scale.

EXPERIENCE

National Telecommunication Institute (NTI) Oct 2026 – Present

Software Engineering & Backend Instructor (Part-Time) Cairo, Egypt

- Deliver an intensive backend engineering curriculum covering Clean Code fundamentals, SOLID principles, and 16+ GoF Creational, Structural, and Behavioral design patterns.
- Train developers on modern Kotlin for backend systems, emphasizing structured concurrency with Coroutines, JVM interoperability, and null safety.
- Instruct enterprise Spring Boot architecture with Kotlin: Aspect-Oriented Programming (AOP), unit and integration testing using JUnit, Mockito, MockMvc, and Testcontainers.
- Guide production API design and security implementations including REST standards, JWT stateless authentication filters, OAuth2 flows, and centralized validation.

Areeb Technology Jul 2025 – Present

Android Software Engineer Cairo, Egypt

- Led KMP research and validated shared logic with native UI integration under Clean Architecture, building a production-ready cross-platform blueprint adopted by multiple teams.
- Refactored a monolithic codebase into 24 independent modules, cutting build time by 30% and enabling parallel team releases with zero merge conflicts.
- Modernized engineering foundations by migrating to Gradle Version Catalogs and adopting Navigation 3, reducing build time by an additional 15%.
- Strengthened app security and compliance by replacing legacy NDK/JNI cryptography with Google Tink and redesigning encrypted DataStore.
- Cut API latency by 20% and improved type-safe serialization by migrating networking from Retrofit to Ktor.
- Owned critical user flows for the Boxooo and Cadaa apps, including onboarding, KYC verification, biometric authentication, password recovery, push notifications, and full RTL/Arabic localization.

Freelance Feb 2025 – Jul 2025

KMP Software Engineer Remote

- Enabled real-time inventory management across Android tablets and Desktop with an offline-first POS and ERP platform built on Kotlin Multiplatform.
- Kept data flowing reliably through connectivity drops with automatic conflict resolution, via a Room and Supabase sync architecture.
- Implemented secure RPC-based stock transfer and inventory workflows to maintain data consistency across multiple warehouse locations.
- Established CI/CD pipelines with automated testing and deployment for continuous delivery.

EDUCATION

Benha University, Faculty of Computer Science and Artificial Intelligence 2026 – Present

M.Sc. in Computer Science (In Progress) Benha, Egypt

Benha University, Faculty of Science Sep 2019 – Jun 2023

B.Sc. in Computer Science, GPA: 3.24/4.0 Benha, Egypt

- Engineered a machine learning solution to predict diabetes using health metrics and classification models.

SKILLS

Languages and Frameworks : Kotlin, Kotlin Multiplatform, Jetpack Compose, Compose Multiplatform, Spring Boot, Swift, SwiftUI, Coroutines, Flow, StateFlow

Architecture and Patterns : Clean Architecture, GoF Design Patterns, MVVM, MVI, Repository Pattern, Dependency Injection (Hilt, Koin), Modular Architecture

Data and Networking : Room, DataStore, Ktor, Retrofit, REST, GraphQL, Supabase, Firebase, SQLite, Protobuf, JWT, OAuth2

Testing and Quality : JUnit, Mockito, Mokkery, Testcontainers, Compose UI Testing, Screenshot Testing, Maestro, Espresso, TDD

DevOps and Tools : Git, GitHub, GitHub Actions, CI/CD, Gradle, Gradle Version Catalogs, ProGuard/R8, Benchmarking

Security and Performance : Google Tink, Encrypted DataStore, Memory Optimization, App Startup Optimization, Performance Profiling

PROJECTS

Safar | *Kotlin Multiplatform, Compose Multiplatform, Supabase*

- Built a delivery operations platform for Android and Desktop that digitizes driver dispatch, invoicing, and payroll, replacing manual, spreadsheet-based operations.
- Developed Supabase Edge Functions for automated payroll logic and implemented Row-Level Security for granular access control.
- Designed the platform to reduce manual operational overhead by 90% through automated invoice tracking and live driver management.

hijri-date-picker | *Kotlin Multiplatform* | Maven Central

- Developed a KMP Hijri calendar picker supporting Android, iOS, and Desktop with platform-consistent behavior.
- Shipped a reusable cross-platform library published to Maven Central with comprehensive documentation and sample apps.
- Engineered custom rendering logic for accurate Hijri date calculations with timezone awareness and localization support.

CrossCart | *Kotlin Multiplatform, Jetpack Compose, Supabase*

- Built an e-commerce platform that lets merchants run a complete storefront (catalog, cart, payments, and order tracking) on Android, iOS, and Desktop from a single codebase.
- Applied clean MVI architecture with unidirectional data flow and live data synchronization via Supabase.
- Designed responsive UI components adapting to different screen sizes and form factors across mobile and desktop platforms.
- Integrated payment processing, cart management, and order tracking with offline support so sales continue even without connectivity.

CERTIFICATIONS

Building Industry-Level Compose Multiplatform Apps, PL-Coding, 2026
Android Developer Track, Digital Egypt Builders Initiative (DEBI), 2025
Android Application Developer Track, ITI, 2023
Advanced Android Nanodegree, Udacity and FWD, 2022
Machine Learning Specialization, DeepLearning.AI (Coursera), 2022
CS50: Computer Science, Harvard University, 2021
Algorithmic Toolbox, Coursera, 2021